1 Lecture 1: Neuromorphic Engineering

1.1 organisation

potentially can get credits for this course but I need to contact the board of examiners asap.

exam will involve knowing the equations for a first questions

activty = mean firing rate.

2 Tutorial one

2.1 1

insert drawing of nmos

2.2 2

equation describing transistor in subthreshold region. in terms of I_r and I_f

I = If - Ir where U_t is the thermal voltage

$I_{ds} = I_0 e^{\frac{\kappa V_g}{V_c}} \left(e^{\frac{-V_s}{U_t}} - e^{\frac{-V_d}{U_t}} \right)$ (should have this equation memorised. full derivation can be found in the book (where???))

2.3 3

what is the effect of changing the source and draing voltage on the forward and reverse currents.

1. increasing $V_s \rightarrow$ decrease I_ds 2. V_d increacing $I_ds \rightarrow$ increase I_ds

$$\frac{I_f}{I_r} = \frac{e^{-V_s/U_t}}{e^{-V_d/U_t}} = e^{V_{ds}/U_t} \tag{1}$$

what happens in the ohmic and saturation regions? $\rightarrow$

the ohmic region is that $V_{ds}(lessthanorequalto)U_t$ saturation region: $V_{ds} > U_t$

2.4 4

$V_{ds}V_d - V_s$

thus in the ohmic region $\frac{I_f}{I_r} \approx 1$ and in the saturation region $\frac{I_F}{I_r} >> 1$

2.5 5

how are the ohmic and saturation regions defined in terms of V_ds? write down approximate current equations.

$$I_{ds} = I_0 e^{\frac{\kappa V_g - V_s}{U_t}(1 - e^{-V_ds/U_t})} \tag{2}$$

Sat: $V_d s > nU_t$ n??? not sure what he is writing on the board there. maybe its kappa

$$I_d s \approx I_0 e^{\frac{\kappa V_g - V_s}{U_t}} \tag{3}$$

ohmic region $e^x = 1 + x + ...$

$$I_{ds} = I_0 e^{\frac{\kappa V_g - V_s}{U_t}} \left(\frac{V_{ds}}{U_t}\right) \tag{4}$$

Vs(Vg)

$$\frac{I_d s}{I_0} = e^{\frac{\kappa V_g - V_s}{U_t}} \tag{5}$$

$$U_t ln(\frac{I}{I_0}) = \kappa V_g - V_s \tag{6}$$

$$V_s = \kappa V_g - U_t \ln\left(\frac{I_{ds}}{I_0}\right) \tag{7}$$

2.6 ...

insert image of pmos.

for both nmos and pmos little arrow is on the source side.

for pmos ground is the drain

for nmos ground is the source